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M1 MACRO

Group B

Session 15

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

$$\underbrace{\begin{pmatrix} \vec{x}_{t+1} \\ \vec{x}_t \end{pmatrix}}_{\vec{x}_{t+1}} = A \underbrace{\begin{pmatrix} \vec{x}_t \\ \vec{x}_{t-1} \end{pmatrix}}_{\vec{x}_t} = A^t \begin{pmatrix} \vec{x}_1 \\ \vec{x}_0 \end{pmatrix}$$

Not given

given

A is "well behaved":

$$\exists P \text{ st } PAP^{-1} = \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

where λ_1, λ_2 are the 2 eigenvalues of A

Result : $y \cdot [ax^2 + bx + c = 0]$ $\lambda_1, \lambda_2 \neq 0$
 $\rightarrow$ roots λ_1, λ_2

then $a + b\mu + c\mu^2 = 0$

$\rightarrow$ roots : $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}$

Proof
 $(\mu \neq 0)$

$$a + b\mu + c\mu^2 = 0$$

change of variable : $\lambda = \frac{1}{\mu} \Rightarrow a + \frac{b}{\lambda} + \frac{c}{\lambda^2} = 0$
 $\Rightarrow \lambda^2 [a\lambda^2 + b\lambda + c] = 0 \Rightarrow$ roots λ_1, λ_2 .

$$0 = \lambda^2 + b\lambda + c \Leftrightarrow a(\lambda - \lambda_1)(\lambda - \lambda_2) = 0$$

$$0 = a + b\lambda + c\lambda^2 \Leftrightarrow c\left(\lambda - \frac{1}{\lambda_1}\right)\left(\lambda - \frac{1}{\lambda_2}\right) = 0$$

$$\Leftrightarrow c, \lambda_1, \lambda_2 \underbrace{(\lambda\lambda_1 - 1)(\lambda\lambda_2 - 1)} = 0$$

$$\Leftrightarrow (1 - \lambda\lambda_1)(1 - \lambda\lambda_2) = 0$$

$$\vec{X}_{t+1} = A \vec{X}_t$$

$$\Leftrightarrow P^{-1} \vec{X}_{t+1} = P^{-1} A \vec{X}_t$$

$$\underbrace{P^{-1} \vec{X}_{t+1}}_{\vec{Z}_{t+1}} = \underbrace{P^{-1} A P}_{\Lambda} \underbrace{P^{-1} \vec{X}_t}_{\vec{Z}_t}$$

$$\vec{Z}_{t+1} = \Lambda \vec{Z}_t$$

$$\Leftrightarrow \begin{bmatrix} z_{t+1}^1 \\ z_{t+1}^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} z_t^1 \\ z_t^2 \end{bmatrix} = \begin{bmatrix} \lambda_1^t & 0 \\ 0 & \lambda_2^t \end{bmatrix} \begin{bmatrix} z_0^1 \\ z_0^2 \end{bmatrix}$$

A

$$\det(A - \lambda I) = 0$$

↳ solutions: λ_1, λ_2

eigenvalues

$$\lambda_1 \rightarrow v_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$A v_1 = \lambda_1 v_1$$

$$Z = P^{-1} X \Leftrightarrow X = P Z$$

$$\Leftrightarrow \begin{bmatrix} \hat{L}_1 \\ \hat{L}_{l-1} \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} Z_1^1 \\ Z_+^2 \end{bmatrix}$$

~~I~~ n period 1:

$$\hat{L}_1 = P_{11} Z_1^1 + P_{12} Z_1^2$$

$$\hat{L}_0 = P_{21} Z_0^1 + P_{22} Z_0^2$$

What is known?

→ I can find $Z_0^1, \hat{L}_1$

$P_{11}, P_{12}, P_{21}, P_{22}$
 $\hat{L}_0$
 $Z_0^2 = 0$ (Transv. Condition)